

TOWA SHIXUN HUANG

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RESEARCH INTERESTS

I am interested in graphics and scientific computing, with a focus on developing efficient and reliable methods for physics-based simulation. My research interests include reduced-order modeling, GPU-accelerated simulation, numerical optimization, and learning-based methods informed by mathematical structure, PDEs, geometry, and symmetry. More broadly, I aim to develop robust and generalizable algorithms for large-scale simulation and numerical computation in graphics, robotics, and scientific applications.

EDUCATION

University of British Columbia

Ph.D. in Computer Science

Vancouver, British Columbia

Sep. 2026 – Jun. 2031

- Advisor: **Peter Yichen Chen**

University of Toronto

B.Sc. in Mathematics and Computer Science

Toronto, Ontario

Sep. 2022 – Jun. 2026

- Advisors: **Eitan Grinspun, Yue Chang**

EXPERIENCE

Dynamic Graphics Project Lab

Apr. 2025 – Aug. 2026

UTEA Undergraduate Student Researcher | **Supervisors:** Eitan Grinspun, Yue Chang

Toronto, ON

- Developed a convex-inspired reduced-order model for robust **physics-based deformable simulation**, maintaining stable behavior under large deformations, contact, and collisions
- Implemented and optimized Newton-based nonlinear solvers and reduced simulation pipelines on both **CPU and GPU**, integrating PCA, autoencoder, and odd-DC subspaces for deformation simulation
- Built **GPU-accelerated synthetic data generation** and benchmarking pipelines with **NVIDIA Warp**, adapting legacy Warp-based simulation scripts and evaluating robustness on challenging out-of-distribution trajectories

Dalla Lana School of Public Health

Jun. 2024 – Jan. 2026

Undergraduate Student Researcher | Jude Dzevela Kong

Toronto, ON

- Developed advanced SEIRS models using differential equations with seasonal forcing via Fourier series, and Bayesian inference (**MCMC, rstan**) for precise parameter estimation calibrated on Canadian influenza data
- Applied numerical methods and nonlinear optimization techniques to refine model calibration and validation

PUBLICATIONS

- *WorldContact: A Contact-Centric World Model for Scalable Robot Learning* [arXiv]
Wang, C., Wang, M., Zhang, H., **Huang, S.**, et al. (2026) | *arXiv preprint*
- *Closing Trajectories: Equation-Free Cyclic Animation via Koopman Surrogates* [CGF]
Huang, S., Chen, S., Chang, Y., Wang, Z., & Chen, P. Y. (2026) | *ACM SIGGRAPH/Eurographics SCA*
- *Odd-DC: Generalizable Neural Model Reduction via Odd Difference-of-Convex Structure* [arXiv]
Huang, S., Grinspun, E., & Chang, Y. (2026) | *Conditionally accepted to ACM TOG*, pending major revision
- *A Systematic Review of Mathematical and Machine Learning Models of Avian Influenza* [DOI]
Huang, S., Bragazzi, N. L., et al. (2025) | *One Health*, 21, 101203

PROJECTS

Superconductor Transition Temperature Prediction | *PyTorch, Transformer* **Nov. 2024 – Dec. 2024**

- Curated superconducting datasets via Materials Project API, including CIF structures and transition temperatures
- Encoded atomic data into 25D Gaussian-mapped feature vectors and designed sentence-structured representations
- Trained Transformer-based models with preliminary MAEs of 0.85–0.99 across two configurations

Numerical Methods for Option Pricing | *Option Pricing, Iterative Methods, Finite Difference* **Mar. 2025 – Apr. 2025**

- Implemented finite difference solvers for the **Black–Scholes PDE** and analyzed numerical stability and convergence
- Solved American put options via projected SOR to enforce early-exercise inequality constraints
- Generated technical visualizations of solution surfaces and convergence trends for reporting

LoRA Rank Scaling Analysis | *PyTorch, Hugging Face, Qwen, LoRA* **Oct. 2025 – Nov. 2025**

- Reviewed key LoRA literature and analyzed intruder-dimension findings through empirical study
- Built Qwen-0.5B fine-tuning pipeline to test LoRA ranks 2–256 and analyze weight matrices on Nvidia GPU
- Found $r = 2$ offers strongest generalization in empirical analysis, while higher ranks overfit via intruder dimensions

AWARDS AND TALKS

Awards: University of Toronto Excellence Award (\$8,000, \$12,000), Dean’s List Scholar (4 years),

Talks: Annual Ontario-Quebec pre-SIGGRAPH Workshop 2025 Speaker, ARIA 2025 Poster Presenter

TEACHING

MAT186: Calculus I, University of Toronto — Teaching Assistant, Fall 2025

MAT187: Calculus II, University of Toronto — Teaching Assistant, Winter 2026

COURSES

Real Analysis; Linear Algebra; Probability and Statistics; Numerical Methods; Differential Equations; Computational Finance (Graduate); Computational Differential Equations (Graduate); Mathematical Theory of Finance (Graduate); Deep Learning Theory (Graduate); Physics-Based Animation

TECHNICAL SKILLS

Technical Topics: Neural Fields, Deep Learning for Animation, Physics-Informed Neural Networks, Graph Neural Networks, Numerical Analysis, LLM Training/Inferencing

Programming Languages: Python, Java, C, C++, MATLAB, R, SQL, Mathematica

Developer Tools: VS Code, Git, Linux Shell, PyCharm, IntelliJ, Blender, Unity, $\LaTeX$

Frameworks and Libraries: pandas, NumPy, SciPy, PyTorch, pymatgen, scikit-learn, PyMC, Warp